



Direct proportion in context

Task A: Conversions

1) Write the following in order from the shortest to the tallest, using the conversion $1 \text{ m} = 3.3 \text{ feet}$

Images are not to scale



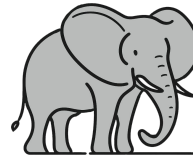
5.1 m



14.2 feet



2.4 m

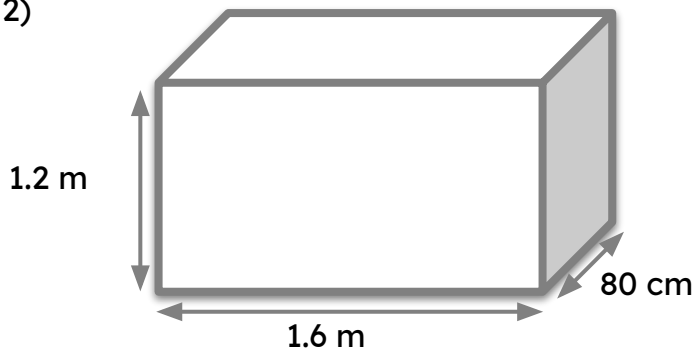


10.2 feet



13.2 feet

2)



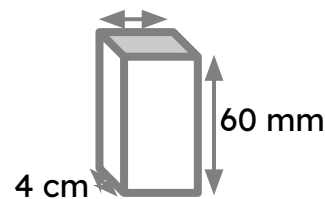
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are not
to scale

Work out the maximum number of the boxes below that will fit into the box above.

a) A cube with dimensions of 40 mm



b) 0.04m



3) Fill in the missing values:

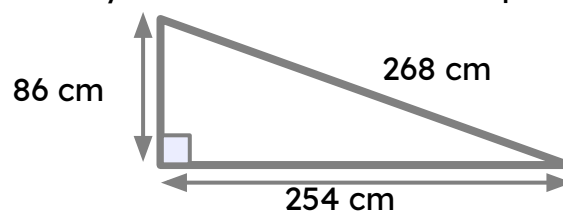
a) $3.2 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

b) $125\,000 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$

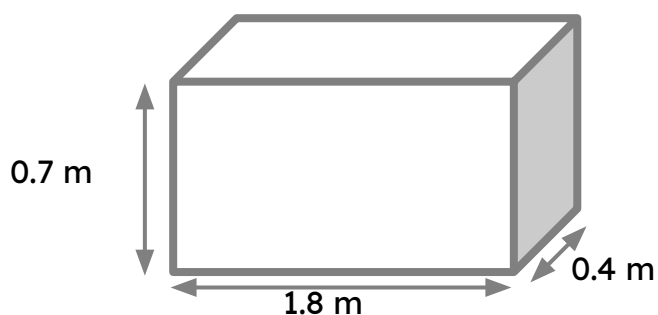
c) $23\,987\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ m}^3$

d) $0.74 \text{ m}^3 = \underline{\hspace{2cm}} \text{ cm}^3$

4) Calculate the area of this triangle. Give your answer to 2 decimal places in m^2 .



5) Calculate the volume of this cuboid. Give your answer in cm^3 .



Task B: Other contexts involving direct proportion

1) Work out the following:

a) 23 pencils cost £14.95 How much will 34 pencils cost?

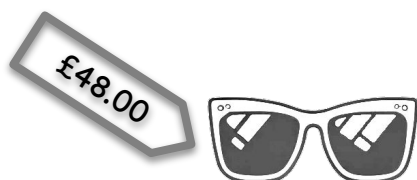
b) 14 rulers cost £5.18 How much will 20 rulers cost?

c) 17 sweets cost £2.55 How much will 37 sweets cost?

d) 27 chocolate bars cost £11.07 How much will 39 chocolate bars cost?

2) Andeep is comparing prices of sunglasses in the UK, the USA and France.

Rank the following from cheapest to most expensive.



3) Calculate the ingredients needed for the following numbers of cookies.

Recipe for 15 cookies
75 g butter
125 g sugar
125 g flour
100 g chocolate chips

a) 18 cookies

b) 72 cookies

c) 6 cookies

d) You have plenty of sugar, flour and chocolate chips, but only 1 kilogram of butter.
What is the largest number of cookies you can make?